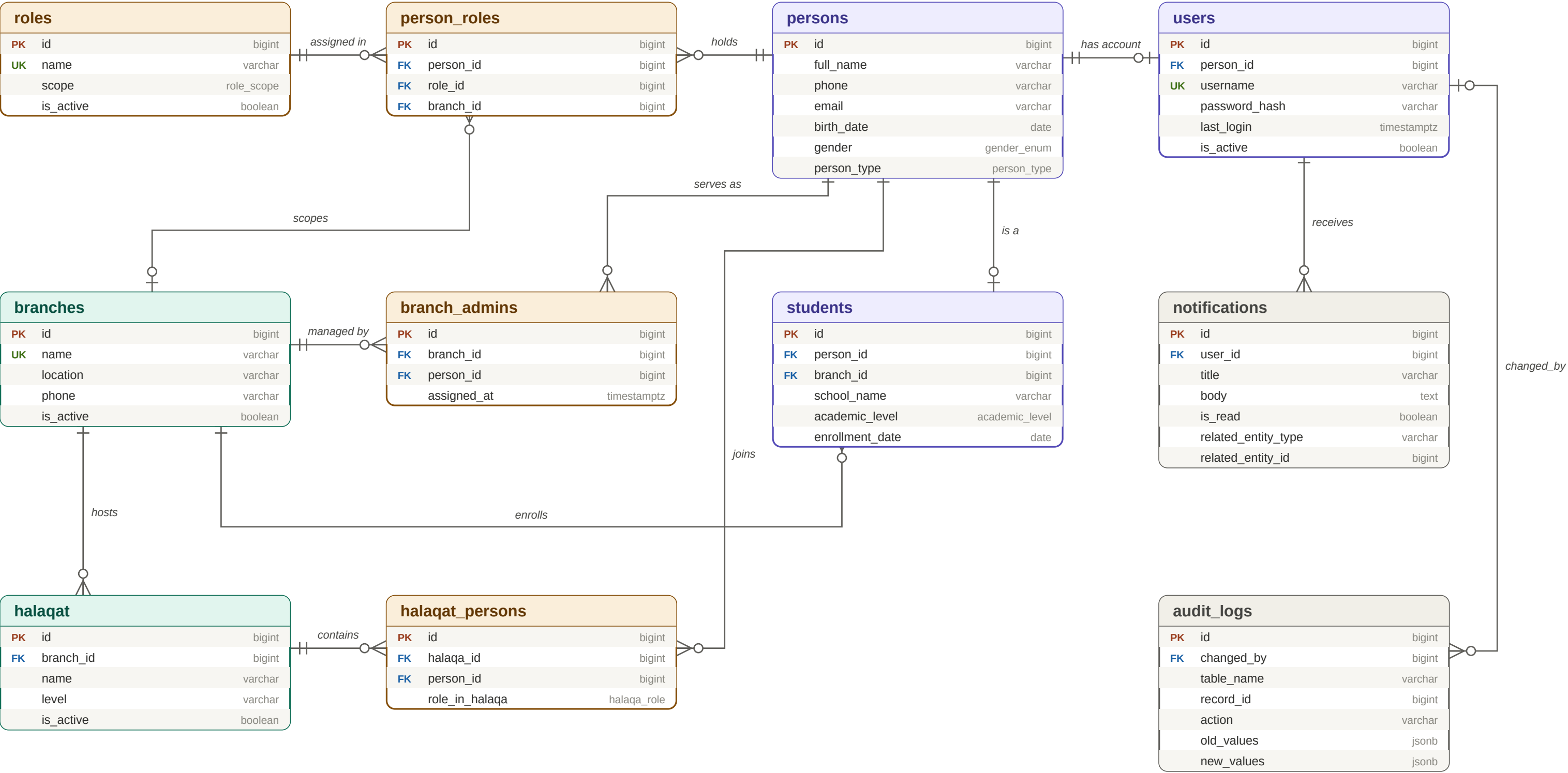


Quran Memorization Association - Database ERD

PostgreSQL 14+ | schema v1.2 | 11 tables, 13 relationships



Legend

- People & accounts
- Organization
- Roles & junction tables
- System
- +

+

 exactly one
- +

○

 zero or one
- ○

 zero or many
- PK

FK

UK

 primary / foreign / unique key

Relationship summary and design notes

Parent	Child	Cardinality	Meaning
roles	person_roles	1 : N	A role can be assigned to many people
persons	person_roles	1 : N	A person can hold several roles
branches	person_roles	0..1 : N	Branch is optional (central roles have none)
persons	users	1 : 0..1	A person has at most one login account
persons	students	1 : 0..1	A person is at most one student record
branches	students	1 : N	Each student belongs to one branch
branches	halaqat	1 : N	A branch hosts many halaqat
branches	branch_admins	1 : N	Max 3 admins per branch (trigger)
persons	branch_admins	1 : N	A person can administer several branches
halaqat	halaqat_persons	1 : N	Membership of a halaqa
persons	halaqat_persons	1 : N	Teacher / student / supervisor, per halaqa
users	notifications	1 : N	Notifications received by a user
users	audit_logs	0..1 : N	changed_by is optional (system changes)

Not drawn as foreign keys

- notifications.related_entity_type / related_entity_id: polymorphic reference, no real FK.
- audit_logs.table_name / record_id: polymorphic reference, no real FK.
- vw_student_search: a view joining students, persons, branches, halaqat and teachers (not a table).

Points worth fixing

- person_roles UNIQUE(person_id, role_id, branch_id): NULL branch_id values are distinct in PostgreSQL, so the same central role can be assigned to one person repeatedly. Use NULLS NOT DISTINCT (PG 15+) or a partial unique index.
- persons.person_type is not tied to rows in students (a 'sheikh' can still have a student row).
- halaqat_persons does not check that a student's branch matches the halaqa's branch.